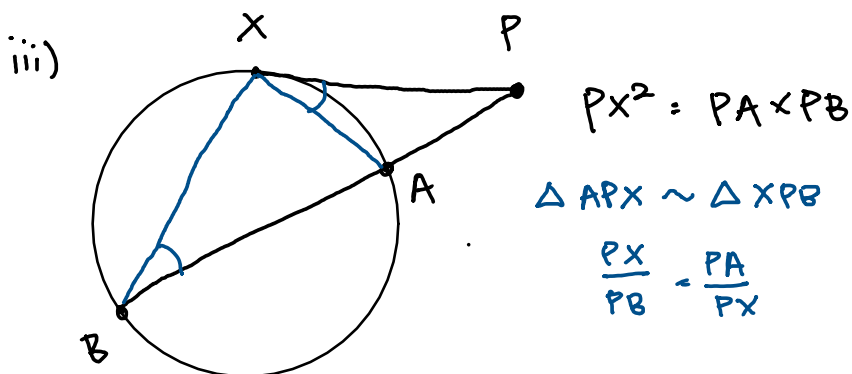
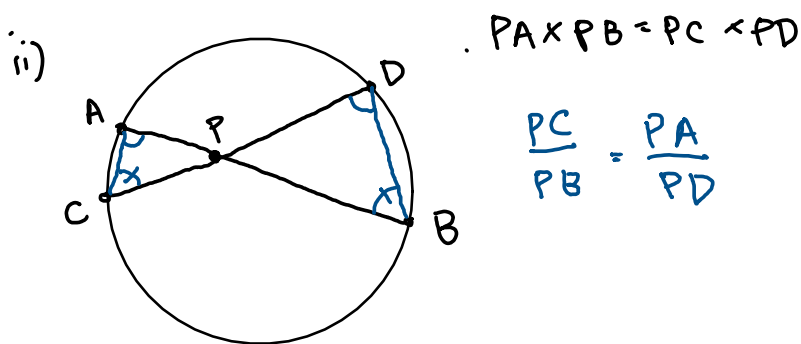
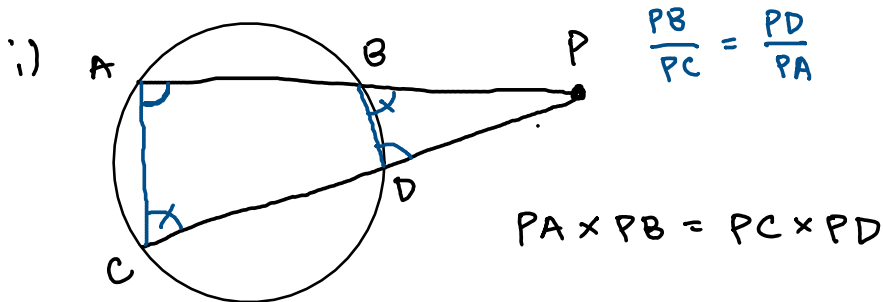


# pop + radax theory

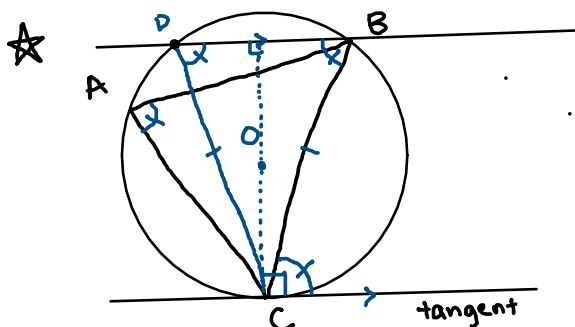
Saturday, 28 June, 2025 10:50 AM

## I: POP THEOREM ( $\Leftrightarrow$ )



a line that passes thru point  
A and B where  $A, B \in \omega$   
and  $A=B$  is a tangent

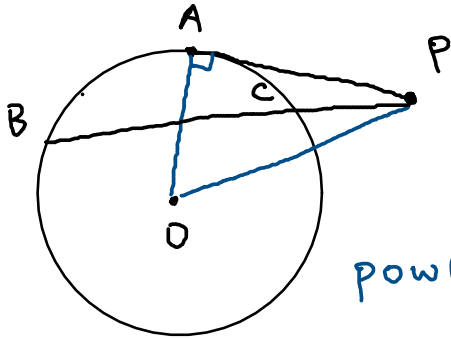
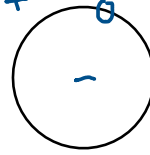
$$\begin{matrix} 2, 2 \\ \uparrow \\ (x-2)^2 \end{matrix}$$



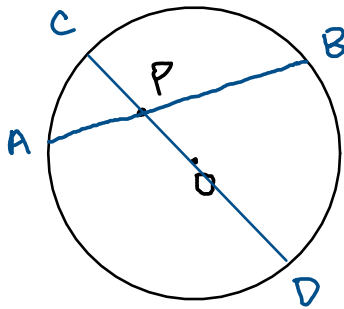
## 2: DEFINING POW

$w$  = circle, center  $O$ , radius  $R$

$$[\text{pow}(P, w) = OP^2 - R^2]$$



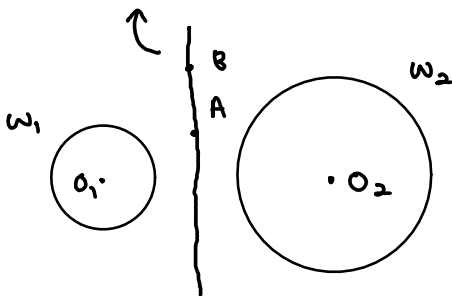
$$\text{pow}(P, w) = PA^2 = PB \times PC$$



$$\begin{aligned} \text{pow}(P, w) &= OP^2 - R^2 \\ &= (OP + R) \times (OP - R) \\ &= PD \times (-PC) \\ &= -(PC \times PD) \\ &= -(PA \times PB) \end{aligned}$$

## 3: RADAX

= radical axis



locus of  $P$  s.t.

$$\text{pow}(P, w_1) = \text{pow}(P, w_2)$$

$$\text{pow}(A, w_1) = \text{pow}(A, w_2)$$

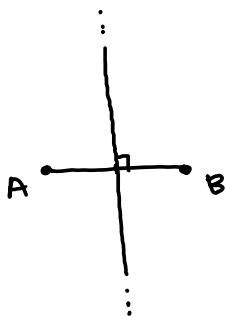
$$\Rightarrow AO_1^2 - R_1^2 = AO_2^2 - R_2^2$$

$$BO_1^2 - R_1^2 = BO_2^2 - R_2^2$$

$$\therefore AO_1^2 - BO_1^2 = AO_2^2 - BO_2^2$$

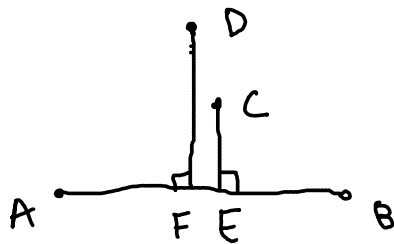
$$AO_1^2 - AO_2^2 = BO_1^2 - BO_2^2$$

$$\Rightarrow AB \perp O_1O_2$$

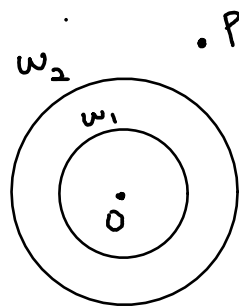
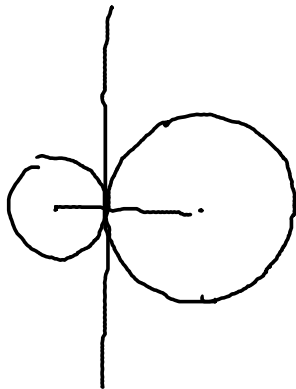
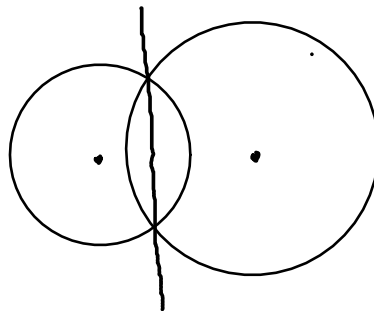
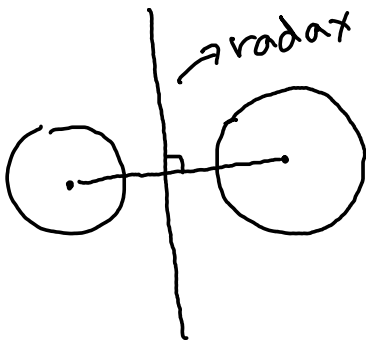


$P$  s.t.  
 $PA = PB$   
locus?

$$AB \perp CD \iff CA^2 - CB^2 = DA^2 - DB^2$$



$$\begin{aligned} CA^2 - CB^2 &= DA^2 - DB^2 \\ \Rightarrow EA^2 - EB^2 &= FA^2 - FB^2 \\ \therefore E &= F \end{aligned}$$

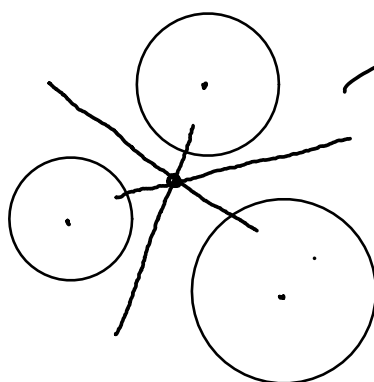


$$\begin{aligned} \text{pow}(P, w_1) &= OP^2 - R_1^2 \\ \text{pow}(P, w_2) &= OP^2 - R_2^2 \end{aligned}$$

$$\text{pow}(P, w_1) \neq \text{pow}(P, w_2)$$

locus is empty

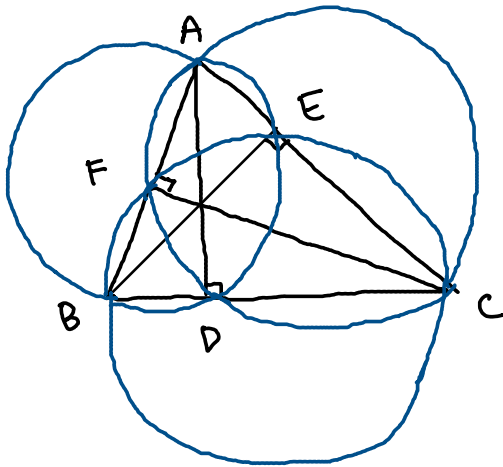
## 4: RADICAL CENTER



radical axis theorem

↳ pairwise radax of 3 non-concentric circles concur

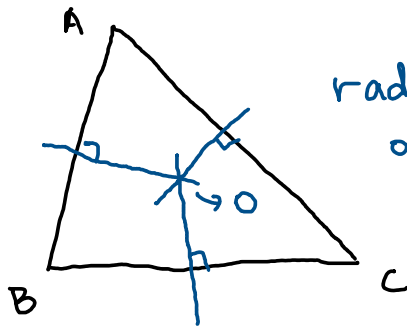
prove the existence  
of the orthocenter,  $H$



## 5 : DEGENERATE CIRCLES

points = circles with radius 0

prove the existence of the circumcenter



radax theorem  
on  $A, B, C$